

Profit Optimization Analysis for Nassau Candy Distributor

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Abstract— This study analyzes transactional sales data from Nassau Candy Distributor to identify profitable products and business divisions. The dataset was cleaned and evaluated using gross margin, profit contribution, revenue contribution, profit per unit, and Pareto analysis. The results indicate that the Chocolate division contributes the highest revenue and profit, while a small number of products account for most of the organization's profitability. Stable margins suggest effective pricing and cost management. The study concludes with practical recommendations for reducing dependency on a limited number of products and improving long-term profitability.

Keywords— Profit Optimization, Pareto Analysis, Business Intelligence, Gross Margin, Data Analytics

I. INTRODUCTION

Profit optimization is an essential objective for distribution companies because revenue alone does not accurately reflect business performance. Companies must identify which products generate sustainable profit, understand cost behavior, and evaluate the financial contribution of different business divisions. Nassau Candy Distributor offers a diverse portfolio of products, making profitability analysis an important decision-support activity. This research evaluates the company's sales data to identify profitable products, examine division-level performance, and provide recommendations that support strategic planning.

II. DATASET AND METHODOLOGY

The study uses transactional sales records containing product information, division names, sales values, costs, gross profit, units sold, and order dates. Prior to analysis, invalid records were removed, missing values were handled, and product names were standardized to improve data quality. Key performance indicators including Gross Margin Percentage, Profit per Unit, Revenue Contribution, Profit Contribution, and Margin Volatility were calculated. Exploratory Data Analysis and Pareto analysis were then applied to identify profitability patterns.

III. RESULTS AND DISCUSSION

The analysis revealed that the Chocolate division consistently outperformed other divisions in terms of revenue and profitability. Products within this category generated strong gross margins, indicating efficient pricing and cost management. Pareto analysis further demonstrated that approximately four products produced nearly eighty percent of the organization's total profit, highlighting a concentration of profitability within a small product group. Although current

financial performance is strong, this concentration introduces operational risk because disruptions affecting these products could significantly influence overall business performance. Margin volatility remained low throughout the study period, suggesting stable pricing strategies and consistent operational efficiency.

IV. RECOMMENDATIONS

The company should prioritize the availability of high-profit products while simultaneously investing in the development of mid-performing products to reduce dependency on a limited portfolio. Continuous monitoring of product margins and expansion of successful pricing strategies across other divisions can improve financial resilience and support sustainable growth.

V. CONCLUSION

The findings demonstrate that profitability analysis provides valuable insights beyond conventional sales reporting. Nassau Candy currently maintains a healthy profit structure supported by strong margins and efficient cost control. However, improving diversification across products and divisions will reduce concentration risk and strengthen long-term business sustainability.

REFERENCES

- [1] Nassau Candy Distributor Sales Dataset (2024–2025).
- [2] Python Documentation
- [3] Streamlit Documentation